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D212 – Task 2

Performance Assessment

Lorraine Ferrusi
DR. KEIONA MIDDLETON

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PART 1: Research Question

A. The Purpose of the Data Mining Report

1.

How can the dimensionality of the dataset be reduced while retaining at least 95% of the variance in the data to simplify the analysis of patient characteristics and outcomes?

2.

The data analysis aims to identify the principal components that capture at least 95% of the variance in the dataset, thereby reducing the dimensionality and simplifying the complexity of the data for a more efficient and insightful analysis of patient characteristics and outcomes.

Once the data set is simplified, more research can be done into the patient characteristics and the health issues to curate better treatment plans that could be individualised to each patient's needs.

PART II: Method Justification

B. Classification Method

1.

PCA (Principal Component Analysis) analyses the dataset by transforming the original variables into a new set of uncorrelated variables called principal components. These components are ordered so that the first few retain most of the variation in the original dataset. (Maćkiewicz & Ratajczak, 1993).

Steps:

- **Standardization:** The data is standardised with a mean of 0 and a standard deviation of 1.
- **Covariance Matrix Computation:** A covariance matrix is computed to understand how dataset variables vary.
- **Eigenvalue and Eigenvector Calculation:** Eigenvalues and eigenvectors of the covariance matrix are calculated. The eigenvectors determine the direction of the new feature space, and the eigenvalues determine their magnitude.
- **Principal Components Selection:** The eigenvectors with the highest eigenvalues are selected as principal components. These components explain the variance in the data the most.
- **Transformation:** The original data is projected onto the selected principal components, reducing the dimensionality.

Expected Outcomes:

- **Explained Variance:** A set of principal components that explain a significant portion of the variance in the dataset, ideally capturing at least 95% of the total variance.
- **Scree Plot:** A plot showing the explained variance by each principal component, helping to visualise the importance of each element.
- **Reduced Dimensionality:** A transformed dataset with fewer dimensions, retaining most of the original information, making it easier to analyse and interpret patient characteristics and outcomes.

2.

PCA relies on several key assumptions to be effective. Some of the assumptions are listed below (Singh, 2022):

- **Linearity:** PCA assumes that the relationships between variables are linear. It captures linear correlations but may not effectively capture non-linear relationships.
- **Mean-Centering:** The data should be mean-centred, meaning each variable should have a mean of zero. This is typically achieved through standardisation.
- **No Multicollinearity:** PCA assumes no perfect multicollinearity among the variables. Highly correlated variables can distort the principal components.

PART III: Data Preparation

C. The Data Preparation Process

1.

The assignment specifies that only continuous variables should be identified. This analysis will include all quantitative fields. PCA can also use categorical variables once the fields are transformed into a suitable format through a one-hot or dummy encoding process (Blaufuks, 2021). However, for this assignment, all non-numerical data was excluded.

Variable	Description	Type	Format	Examples
Lat	GPS coordinates of patient's residence as listed on the billing statement	Continuous	Float	-0.814667639
Lng	GPS coordinates of the patient residence as listed on the billing statement	Continuous	Float	0.2971344413
Population	Population within a mile radius of the patient, based on census data	Numerical	Integer	2951
Children	Household's number of children	Numerical	Integer	1, 3, NA
Age	Patient's age	Numerical	Integer	18, 56, 25
Income	Annual income	Continuous	Float	55586.48
Vit D Levels	ng/mL measure vitamin D levels	Continuous	Float	16.49455
Doc Visits	Number of times the doctor visited the patient	Numerical	Integer	4, 7, 2...
Full Meals Eaten	Patient's number of full meals during stay	Numerical	Integer	0, 1, 2...
VitD_supp	The number of times that vitamin D supplements were administered to the patient	Numerical	Integer	0, 1, 2...
Initial Days	Number of days in hospital for first visit	Continuous	Float	5.95725, 1.012586
Total Charge	Daily costs of patient's stay	Continuous	Float	2990.086, 3316.4
Additional Charges	Other charges during stay	Continuous	Float	18518.04, 10575.43

2.

Uniformity in machine learning algorithms is crucial for enhancing performance (Pal, 2023). This uniformity is achieved through standardisation. In Python, the StandardScaler package is used to standardise the data.

See attached CSV file (df_scaled.csv).

See attached Python file for data cleaning code (D212 Task 2 Data Cleaning.py).

PART IV: Analysis

D. Perform Data Analysis and Report on the Results

1.

To determine the covariance matrix and display it as a heatmap, the following steps can be followed:

```
# initiate pca to get the matrix
```

```
principal_components = pca.fit_transform(data_final)
```

```
# convert to DataFrame for better readability
```

```
principal_components_df = pd.DataFrame(principal_components, columns  
                                       = [f'PC{i + 1}' for i in range(principal_components.shape[1])])
```

```
# compute the covariance matrix of the principal components
```

```
covariance_matrix = np.cov(principal_components_df.T)
```

```
# print the covariance matrix using Pandas for better formatting
```

```
print(pd.DataFrame(covariance_matrix, columns
    = [f'PC{i + 1}' for i in range(principal_components.shape[1])], index
    = [f'PC{i + 1}' for i in range(principal_components.shape[1])]))
```

```

           PC1          PC2    ...          PC12          PC13
PC1  1.995695e+00  1.257786e-16  ... -4.103795e-17 -2.051897e-17
PC2  1.257786e-16  1.715488e+00  ... -1.054484e-16  4.814131e-17
PC3 -1.647736e-16 -4.974297e-17  ...  2.043015e-18 -3.691861e-18
PC4 -3.453583e-16  1.687708e-18  ... -3.304354e-17 -3.752929e-18
PC5 -1.474524e-16 -3.627683e-16  ...  1.540255e-16  3.708516e-18
PC6  5.755972e-17 -1.104116e-16  ...  8.625075e-17  2.170703e-17
PC7 -1.610429e-16 -3.142690e-16  ...  2.409980e-17 -3.569724e-17
PC8 -1.350388e-16  1.134317e-16  ...  4.992062e-17  4.818850e-18
PC9 -3.236846e-16 -1.035720e-16  ... -1.013513e-16  1.043714e-17
PC10 -2.525344e-16 -3.868626e-16  ... -4.041616e-17  2.909075e-17
PC11 -6.568736e-17 -1.527820e-17  ... -2.673684e-17 -6.884071e-19
PC12 -4.103795e-17 -1.054484e-16  ...  2.833950e-01  1.272998e-17
PC13 -2.051897e-17  4.814131e-17  ...  1.272998e-17  1.171295e-02

```

```
# display covariance matrix as a heatmap
```

```
sns.heatmap(covariance_matrix, annot = False, fmt = ".2f", cmap = "coolwarm", xticklabels
    = [f'PC{i + 1}' for i in range(principal_components.shape[1])], yticklabels
    = [f'PC{i + 1}' for i in range(principal_components.shape[1])])
```

```
plt.show()
```

```
# create a scree plot for PCA
```

```
plt.figure(figsize = (10,6))
```

```
plt.plot(range(1, len(explained_variance) + 1), explained_variance, marker = 'o', linestyle = '--')
```

```
plt.title('Scree Plot')
```

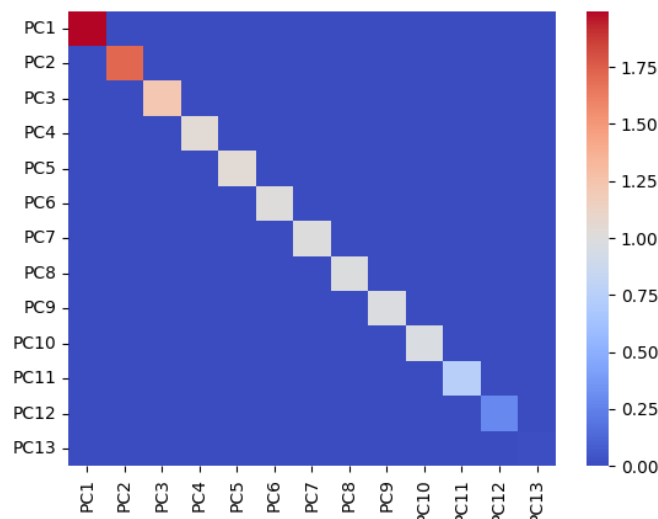
```
plt.xlabel('Principal Component')
```

```
plt.ylabel('Explained Variance Ratio')
```

```
plt.xticks(np.arange(1, len(explained_variance) + 1, step = 1))
```

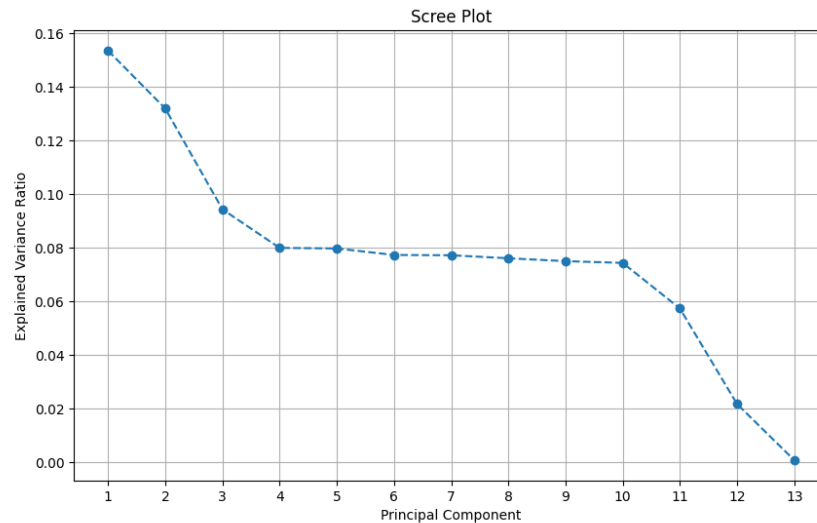
```
plt.grid(True)
```

```
plt.show()
```

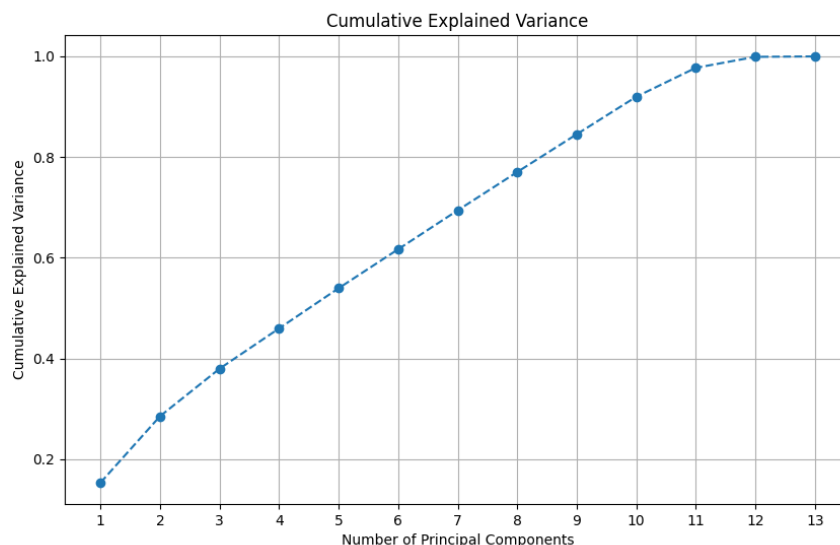


2.

Using the elbow method alone is insufficient to determine the principal components total number. The scree plot shows the explained variance ratio for each principal component. The explained variance ratios are relatively small, indicating that each principal component captures only a small portion of the total variance. This suggests that the data may have many dimensions with relatively equal contributions to the variance, making it difficult for the first few principal components to capture a significant amount of the total variance.



Creating a cumulative explained variance chart along with the elbow method will give the optimal insight into the number of principal components. The scree plot and cumulative explained variance plot indicate that each principal component captures only a small portion of the total variance. The cumulative explained variance plot shows that many principal components are needed to capture a significant portion of the total variance. This suggests that the data has many dimensions with relatively equal contributions to the variance, making it challenging for the first few principal components to capture a substantial amount of the total variance. To choose the number of components to keep, look for the "elbow" point in the cumulative explained variance plot, where the explained variance starts to level off or select a threshold (e.g., 90%) of the cumulative explained variance.



For this analysis, the total number of principal components that would explain is 11.

3.

To print the explained variance for each of the principal components, the following steps are required:

```
#define PCA model to use
pca = PCA(n_components = 11)

#fit PCA model to data
pca_fit = pca.fit(data_final)

# create a DataFrame to display principal components and their explained variance ratios
principal_components_df = pd.DataFrame({
    'Principal Component': [f'PC{i + 1}' for i in range(len(variance_each_pc))],
    'Explained Variance Ratio': variance_each_pc })
print(principal_components_df)
```

	Principal Component	Explained Variance Ratio
0	PC1	0.15349968
1	PC2	0.1319474143
2	PC3	0.0943815101
3	PC4	0.080012544
4	PC5	0.0797879449
5	PC6	0.0773755228
6	PC7	0.0772561044
7	PC8	0.0761362552
8	PC9	0.0750677907
9	PC10	0.0744119171
10	PC11	0.0574249772

4.

The total variance is calculated by adding the explained variance ratios

```
# sum the explained variance ratios of the first 11 principal components
variance_11_pcs = np.sum(explained_variance[:11])
print(variance_11_pcs)
```

5.

The PCA analysis on the scaled dataset revealed that the first 11 principal components capture approximately 97.73% of the total variance. The explained variance ratios for these components range from 15.35% for PC1 to 5.74% for PC11. The top contributing original features for the first few principal components include 'TotalCharge', 'Initial_days', 'Additional_charges', 'Age', 'Lat', and 'Population', indicating that these features play a significant role in capturing the variance in the dataset.

See attached Python file (D212 Task 2 PCA.py)

References

- Blaufuks, W. (2021). *FAMD: How to generalize PCA to categorical and numerical data*. Retrieved from Medium: <https://towardsdatascience.com/famd-how-to-generalize-pca-to-categorical-and-numerical-data-2ddb2b9210>
- Maćkiewicz, A., & Ratajczak, W. (1993). Principal components analysis (PCA). *Computers & Geosciences*, 303-342.
- Pal, S. (2023). *StandardScaler()*. Retrieved from GeekPython: <https://geekpython.in/how-to-use-sklearn-standardScaler#:~:text=StandardScaler%20is%20used%20to%20standardize,to%20differences%20in%20feature%20scales>.
- Singh, D. (2022). *Assumptions and limitations of PCA*. Retrieved from Kaggle: <https://www.kaggle.com/discussions/general/372597>